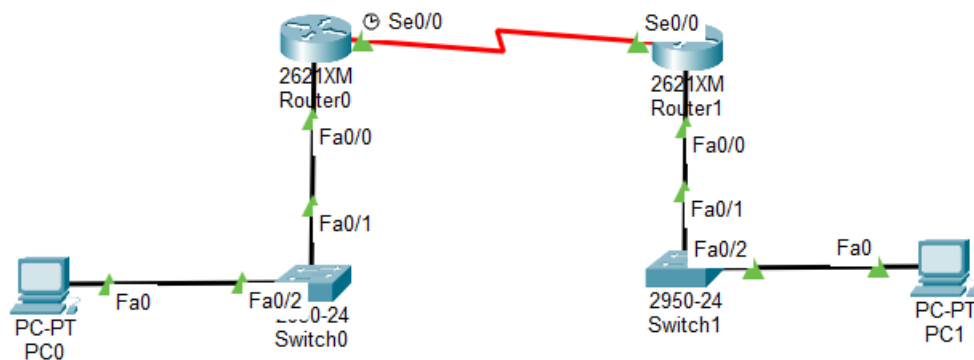


b.WAN Authentication Protocols: Authentication is Needed in WAN, WAN Authentication Protocols: HDLC, PPP Authentication (PAP, CHAP, MS-CHAP).

Task 1.

Draw a simple network diagram showing two routers connected over a WAN link and label where authentication is needed to prevent unauthorized access.



Explanation:

- Two routers communicate through a WAN link.
- Authentication is required on the WAN link to ensure that only an authorized router can establish the connection.
- With PPP, PAP or CHAP can be used for authentication.

Task 2.

**Configure a sample HDLC WAN connection between two routers using Packet Tracer or any network simulator, and explain why HDLC does not provide authentication.

Hint: Focus on the commands used and what is missing in terms of security.**

Router 1

```
Router#show interfaces serial 0/0
Serial0/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.0.0.1/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 output buffer failures, 0 output buffers swapped out
--More--
```

Router 2

```
Router#show interfaces serial 0/0
Serial0/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.0.0.2/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
  DCD=up DSR=up DTR=up RTS=up CTS=up

Router#
```

Why HDLC does not provide authentication

HDLC is a Layer 2 WAN encapsulation protocol that provides framing and basic link communication, but standard Cisco HDLC does not have a built-in username/password authentication mechanism.

If authentication is required, PPP is preferred because PPP supports PAP and CHAP.

Set up a PPP connection between two routers in Packet Tracer and enable PAP (Password Authentication Protocol) authentication. Show the configuration commands and explain what happens if the password is incorrect.

```
hostname R1
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username R2 password 0 cisco
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/0
 ip address 10.0.0.1 255.255.255.252
 encapsulation ppp
 ppp authentication pap
 ppp pap sent-username R1 password 0 cisco
 clock rate 64000
```

```
Router(config)#hostname R2
R2(config)#username R1 password cisco
R2(config)#interface serial 0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#encapsulation ppp
R2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

R2(config-if)#ppp authentication pap
R2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to down

R2(config-if)#ppp pap sent-username R2 password cisco
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#do wr
Building configuration...
[OK]
R2(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
```

What happens if the password is incorrect?

For example, suppose R1 sends:

```
ppp pap sent-username R1 password wrongpassword
```

but R2 expects:

```
R1 / cisco
```

The PAP authentication will fail, and the PPP connection will not successfully establish. The serial link may show authentication-related errors and communication across the WAN will fail.

Important point about PAP

PAP sends the username and password as part of the authentication exchange in a relatively simple manner. Therefore, PAP provides weaker security than CHAP.

Task 4.

Modify your PPP configuration to use CHAP (Challenge Handshake Authentication Protocol) instead of PAP, and describe one key difference in how CHAP improves security over PAP.

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#username R2 password cisco
R1(config)#do wr
Building configuration...
[OK]
R1(config)#interface serial 0/0
R1(config-if)#encapsulation ppp
R1(config-if)#ppp authentication chap
R1(config-if)#no shutdown
R1(config-if)#exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to down

R1(config)#do wr
Building configuration...
[OK]
R1(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
```

```
R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#hostname R2
R2(config)#username R1 password cisco
R2(config)#do wr
Building configuration...
[OK]
R2(config)#interface serial 0/0
R2(config-if)#encapsulation ppp
R2(config-if)#ppp authentication chap
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#dow r
^
```

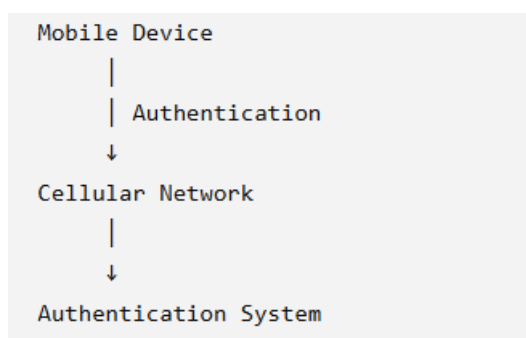
Feature	PAP	CHAP
Authentication	Username/password	Challenge-response
Password sent directly	Yes	No
Security	Lower	Higher
Authentication process	Simple	Challenge and response
Packet Tracer command	ppp authentication pap	ppp authentication chap

Task 5.

Research and list two real-world scenarios (such as public WiFi login or mobile data authentication) where WAN authentication protocols like PAP, CHAP, or MS-CHAP are used, and explain why authentication is critical in each case.

Scenario 1 — Mobile data / cellular networks

Mobile networks use authentication mechanisms to verify that a subscriber is authorized to use the network.

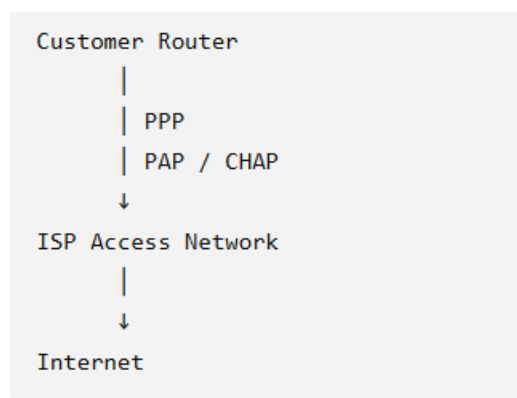


Why authentication is critical:

- It prevents unauthorized users from accessing network services and helps the operator associate network access with a valid subscriber/account.
- Modern cellular networks generally use specialized cellular authentication systems rather than plain PAP or CHAP; PAP/CHAP are more directly associated with older or particular packet-data/PPP arrangements.

Scenario 2 — Public Wi-Fi / ISP access

Some ISP or managed Wi-Fi systems use PPP-based authentication, including PAP or CHAP, between a customer's equipment and the provider's access network.



Why authentication is critical:

- The ISP needs to verify that the customer is authorized to use the service.
- Authentication also helps prevent unauthorized access and allows the provider to associate a connection with the correct subscriber.